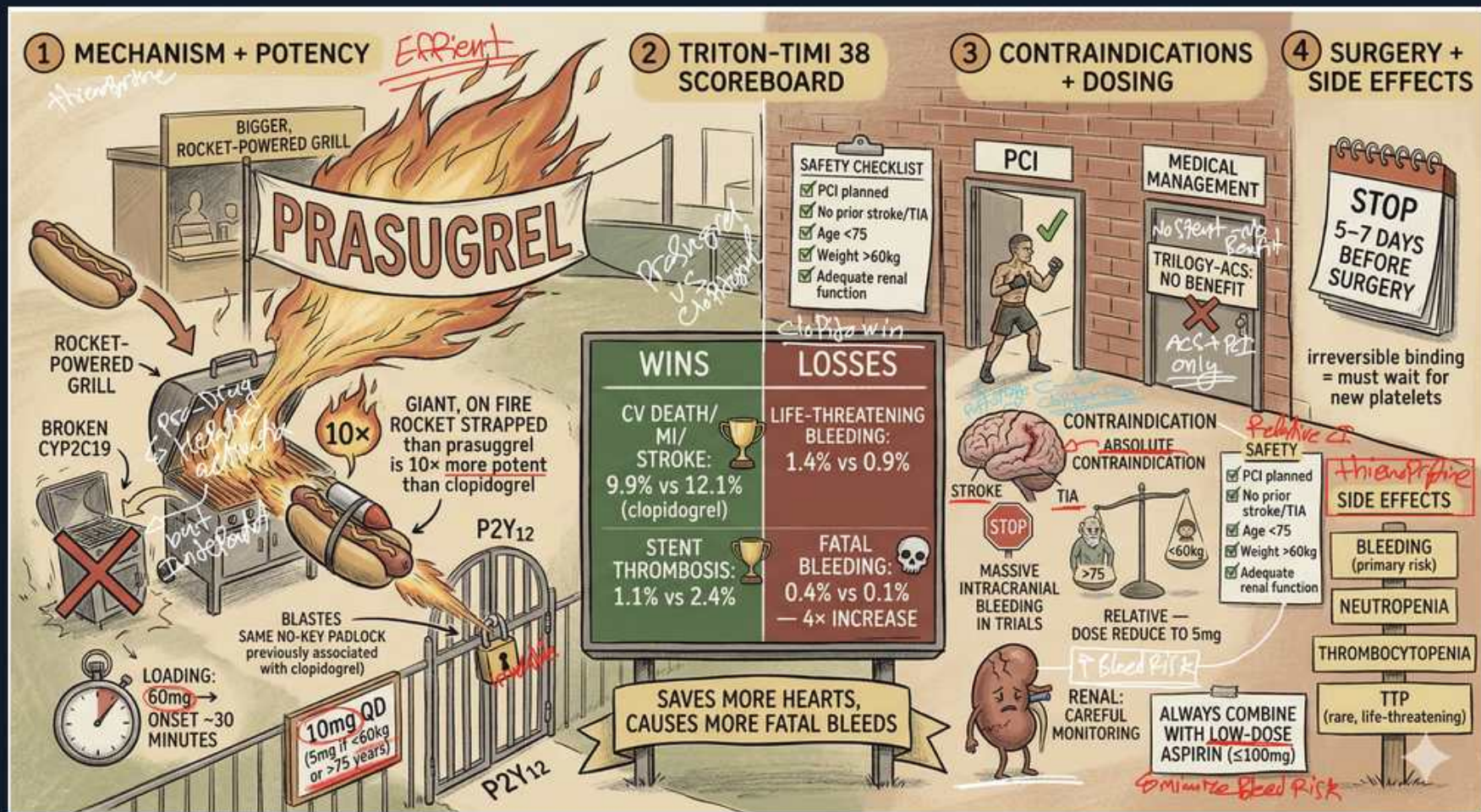


Antiplatelets — Prasugrel (Rocket-Powered Grill, TRITON-TIMI 38)



1. Prasugrel — rocket-powered potency

WHAT YOU SEE

A giant flaming rocket-powered grill labeled PRASUGREL with '10x', 'bigger, rocket-powered grill'.

WHAT IT ENCODES

Prasugrel is an irreversible thienopyridine P2Y12 inhibitor that is ~10x more potent than clopidogrel, giving faster, stronger, more consistent platelet inhibition. Same target (P2Y12) as clopidogrel.

BOARD TRAP

Greater potency = more bleeding, not a free win. Prasugrel saves more ischemic events but causes more major/fatal bleeding than clopidogrel.

2. Efficient CYP activation (one-step)

WHAT YOU SEE

Broken CYP2C19 grill with red X; note 'efficient' / no-key padlock; prodrug still but activated reliably.

WHAT IT ENCODES

Prasugrel is also a prodrug but is activated efficiently in essentially one CYP step and is NOT dependent on CYP2C19, so it bypasses the poor-metabolizer problem and PPI interaction.

BOARD TRAP

Unlike clopidogrel, prasugrel response is NOT reduced by CYP2C19 loss-of-function or omeprazole. Choose prasugrel/ticagrelor over clopidogrel in CYP2C19 poor metabolizers.

3. Irreversible P2Y12 blockade

WHAT YOU SEE

Rocket 'blasts the no-key padlock' at the P2Y12 receptor, P2Y12 labels around.

WHAT IT ENCODES

Like clopidogrel, prasugrel irreversibly inhibits P2Y12 for the platelet lifespan, so its effect persists after the drug clears and requires new platelets to recover.

BOARD TRAP

Irreversible -> hold 5-7 days before surgery. Don't expect rapid offset just because the plasma drug clears.

4. Loading 60 mg, onset ~30 min

WHAT YOU SEE

Stopwatch 'LOADING 60mg, onset ~30 minutes'; maintenance '10 mg QD (5 mg if >=75 yrs / <60 kg)'.

WHAT IT ENCODES

Prasugrel loads 60 mg with rapid ~30 min onset; maintenance is 10 mg daily, reduced to 5 mg in the elderly (>=75) or low body weight (<60 kg) to limit bleeding.

BOARD TRAP

The 5 mg reduced maintenance dose is specifically for age >=75 OR weight <60 kg. Giving full 10 mg to a frail elderly/low-weight patient is the dosing trap.

5. TRITON-TIMI 38 scoreboard

WHAT YOU SEE

A stadium 'TRITON-TIMI 38 SCOREBOARD' with WINS and LOSSES columns.

WHAT IT ENCODES

TRITON-TIMI 38 compared prasugrel vs clopidogrel in ACS patients going to PCI. Prasugrel reduced ischemic events but increased major bleeding — the trial that defined its niche and contraindications.

BOARD TRAP

Prasugrel's benefit was in ACS patients with PLANNED PCI/known coronary anatomy. It is not for medically managed ACS (TRILOGY-ACS showed no benefit there).

6. WINS — fewer MI / stent thrombosis

WHAT YOU SEE

Green WINS column: CV death/MI/stroke 9.9% vs 12.1% (clopidogrel); stent thrombosis 1.1% vs 2.4%.

WHAT IT ENCODES

Prasugrel cut the composite of CV death/MI/stroke (9.9% vs 12.1%) and roughly halved stent thrombosis (1.1% vs 2.4%) compared with clopidogrel. 'Saves more hearts.'

BOARD TRAP

The ischemic benefit is real but must be weighed against bleeding — net clinical benefit was lost or harmful in the contraindicated subgroups (prior stroke/TIA, elderly, low weight).

7. LOSSES — more fatal bleeding

WHAT YOU SEE

Red LOSSES column: life-threatening bleeding 1.4% vs 0.9%; fatal bleeding 0.4% vs 0.1% (4x increase).

WHAT IT ENCODES

Prasugrel increased life-threatening bleeding (1.4% vs 0.9%) and fatal bleeding (0.4% vs 0.1%, ~4x). 'Causes more fatal bleeds' — the cost of greater potency.

BOARD TRAP

The ~4x fatal-bleed increase drives the contraindications. A board stem with prior stroke/TIA on prasugrel is testing net harm from bleeding.

8. Absolute contraindication — prior stroke/TIA

WHAT YOU SEE

Brain with STOP sign: 'ABSOLUTE CONTRAINDICATION — prior stroke/TIA -> massive intracranial bleeding in trials'.

WHAT IT ENCODES

Prasugrel is ABSOLUTELY contraindicated with any history of stroke or TIA because of net harm from intracranial bleeding observed in TRITON-TIMI 38. This is its hardest rule.

BOARD TRAP

Prior stroke/TIA = do NOT use prasugrel (absolute). This is the single most tested prasugrel contraindication; choose clopidogrel or ticagrelor instead.

9. Relative caution — age ≥ 75 , weight < 60 kg

WHAT YOU SEE

'RELATIVE — age ≥ 75 ; weight < 60 kg -> dose reduce to 5mg'; kidney 'renal: careful monitoring'.

WHAT IT ENCODES

Age ≥ 75 and weight < 60 kg are RELATIVE cautions (higher bleed risk); if used, reduce maintenance to 5 mg daily. Renal impairment warrants careful monitoring.

BOARD TRAP

These are RELATIVE (dose-reduce to 5 mg), versus prior stroke/TIA which is ABSOLUTE (avoid entirely). Mixing relative vs absolute contraindications is the classic trap.

10. PCI checklist — when prasugrel fits

WHAT YOU SEE

Safety checklist: PCI planned, no prior stroke/TIA, age < 75 , weight > 60 kg, adequate renal function; PCI room thumbs-up.

WHAT IT ENCODES

Ideal prasugrel candidate: ACS with PLANNED PCI, no stroke/TIA history, age < 75 , weight ≥ 60 kg, adequate renal function. Outside this profile, prefer an alternative.

BOARD TRAP

All checklist boxes must be satisfied. Failing any (stroke/TIA, elderly, low weight) flips prasugrel from first-line to contraindicated/dose-reduced.

11. Medical management — no benefit (TRILOGY-ACS)

WHAT YOU SEE

Medical Management panel: 'TRILOGY-ACS = NO BENEFIT', 'ACS+PCI only'.

WHAT IT ENCODES

In medically managed ACS without PCI (TRILOGY-ACS), prasugrel showed NO benefit over clopidogrel. Its role is specifically ACS undergoing PCI with known anatomy.

BOARD TRAP

Don't start prasugrel in conservatively (non-PCI) managed ACS — no benefit there. Ticagrelor, by contrast, is approved across ACS including medical management.

12. Stop 5-7 days before surgery

WHAT YOU SEE

A STOP sign: 'STOP 5-7 DAYS BEFORE SURGERY — irreversible binding -> must wait for new platelets'.

WHAT IT ENCODES

Prasugrel must be stopped ~7 days before elective surgery (longer than clopidogrel's 5) because of strong irreversible inhibition; recovery requires regenerating fresh platelets.

BOARD TRAP

Prasugrel needs ~7 days off (vs ~5 for clopidogrel/ticagrelor). Using the clopidogrel hold time for prasugrel under-doses the washout.

13. Side effects — bleeding, TTP, cytopenias

WHAT YOU SEE

SIDE EFFECTS strip: bleeding (primary risk), neutropenia, thrombocytopenia, TTP (rare, life-threatening); 'fatal-bleed risk'.

WHAT IT ENCODES

Primary adverse effect is bleeding; rare thienopyridine-class effects include neutropenia, thrombocytopenia, and TTP. Bleeding dominates prasugrel's risk profile.

BOARD TRAP

Bleeding (not dyspnea) is prasugrel's signature toxicity. Dyspnea and ventricular pauses point to TICAGRELOR instead — don't attribute those to prasugrel.

14. Always combine with low-dose aspirin

WHAT YOU SEE

Banner 'ALWAYS COMBINE WITH LOW-DOSE ASPIRIN (≤ 100 mg)'.

WHAT IT ENCODES

Prasugrel is used as DAPT with low-dose aspirin (≤ 100 mg). Low-dose aspirin is mandated in combination to keep bleeding manageable while covering both TXA2 and P2Y12 pathways.

BOARD TRAP

Keep aspirin ≤ 100 mg in any DAPT/triple combo. For ticagrelor specifically, higher aspirin doses (> 100 mg) reduce its efficacy — a paired board point.
